Coding / Audit

PDF edition of preserved repository document | docs/coding/audit.md

Historical statements and test counts below retain their original scope. Consult the master report for current status.

Flagship transition audit

The multiplication release mindscape-final-study-v1 remains immutable. Its 92-test implementation and frozen negative findings are historical/control evidence, not coding results. No old checkpoint or experiment is overwritten.

The generic structural Environment protocol and domain-dispatched runner can be reused. Concrete decimal State/Observation/Action records cannot represent files or edits without inappropriate fields; additive coding dataclasses are necessary. Memory storage and bounded DreamEngine already accept generic payloads. The multiplication implementation requires no changes.

The existing boolean answer evaluator cannot independently execute hidden code tests; a coding backend must bind a final repository hash to independently executed private tests. Learner-facing views must omit hidden tests, correct source and teacher patches. Visible-test feedback is permitted; hidden evaluation is terminal and never returned during the repair loop.

Current hardware: Apple M5, 10 CPUs,16GB. Local CPU PyTorch/Transformers is installed; no Docker/Podman/Ollama executable was found. Native sandbox-exec exists but the actual launch failed sandbox_apply: Operation not permitted. Host Python execution of generated code is therefore forbidden. The planned fallback is WASI Python through Wasmtime, preopening only a disposable task directory and curated Python standard library, with no network or unrelated host filesystem capability. Probe file/network/process denial, timeout and memory limits before any benchmark code executes. Fail closed if the runtime cannot enforce them.

Two Apache2.0 coding backbones (Qwen2.5-Coder0.5B and1.5B Instruct) are candidates. Actual pilot capability and CPU latency, not nominal size, will select the practical backbone. Model-only generation, structured generation and real bounded repair loops must remain distinct. Domain-data adaptation must be labeled precisely; retrieval/sample budgets are not gradient-training budgets. Unequal interaction/supervision/compute remains an explicit comparison limit.